6E STANDARD ENTHALPY CHANGE OF FORMATION AND HESS'S LAW

SPECIFICATION REFERENCE

6.4 PART	6.6	CP2

LEARNING OBJECTIVES

- Be able to define standard enthalpy change of formation.
- Be able to construct enthalpy cycles using Hess's Law.
- Be able to calculate enthalpy changes from data using Hess's Law.

WHAT IS MEANT BY STANDARD ENTHALPY CHANGE OF FORMATION?

The **standard enthalpy change of formation** ($\Delta_f H^\ominus$) is the enthalpy change measured at 100 kPa and a specified temperature, usually 298 K, when one mole of a substance is formed from its elements in their standard states.

For the purposes of this definition, the standard state of an element is the form in which it exists at the specified temperature, usually 298 K, and a pressure of 100 kPa.

The standard enthalpy change of formation of gaseous carbon dioxide is the enthalpy change for the reaction:

$$C(s, \text{graphite}) + O_2(g) \rightarrow CO_2(g) \qquad \Delta_f H^\ominus = -394\,kJ\,mol^{-1}$$

The standard enthalpy change of formation of liquid ethanol is the enthalpy change for the reaction:

$$2C(s, \text{graphite}) + 3H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow C_2H_5OH(l)$$
$$\Delta_f H^\ominus = -278\,kJ\,mol^{-1}$$

LEARNING TIP

Enthalpy changes of formation emphasise the necessity of including state symbols in thermochemical equations. The value for the standard enthalpy change of formation of gaseous ethanol is different from that for liquid ethanol:

$\Delta_f H^\ominus[C_2H_5OH(l)] = -278\,kJ\,mol^{-1}$

$\Delta_f H^\ominus[C_2H_5OH(g)] = -235\,kJ\,mol^{-1}$

Note that the enthalpy change of formation for an element is zero.

HESS'S LAW

Most standard enthalpy changes of formation cannot be determined experimentally. For example, it is impossible to burn carbon in oxygen and form solely carbon monoxide. So, the enthalpy change for the reaction

$$C(s) + \tfrac{1}{2}O_2(g) \rightarrow CO(g)$$

is impossible to determine directly.

Fortunately, we can make use of **Hess's Law**, which is an application of the first law of thermodynamics (the law of conservation of energy).

Hess's Law states that the enthalpy change of a reaction is independent of the path taken in converting reactants into products, provided the initial and final conditions are the same in each case.

EXAM HINT

You should learn the definition of Hess's Law as it is frequently the first part of a question on energetics.

Hess's Law allows us to calculate the standard enthalpy change of formation of carbon monoxide.

The enthalpy changes of combustion of carbon and carbon monoxide can both be determined experimentally. Their values are:

$$C(s, \text{graphite}) + O_2(g) \rightarrow CO_2(g) \qquad \Delta_c H^\ominus = -394\,kJ\,mol^{-1}$$
$$CO(g) + \tfrac{1}{2}O_2(g) \rightarrow CO_2(g) \qquad \Delta_c H^\ominus = -283\,kJ\,mol^{-1}$$

There are two ways to use these data to calculate the standard enthalpy change of formation of carbon monoxide.

METHOD 1: SUBTRACT EQUATIONS

Reverse the second equation above and then add to the first:

$$CO_2(g) \rightarrow CO(g) + \tfrac{1}{2}O_2(g) \qquad \Delta_c H^\ominus = +283\,kJ\,mol^{-1}$$
$$C(s, \text{graphite}) + O_2(g) \rightarrow CO_2(g) \qquad \Delta_c H^\ominus = -394\,kJ\,mol^{-1}$$

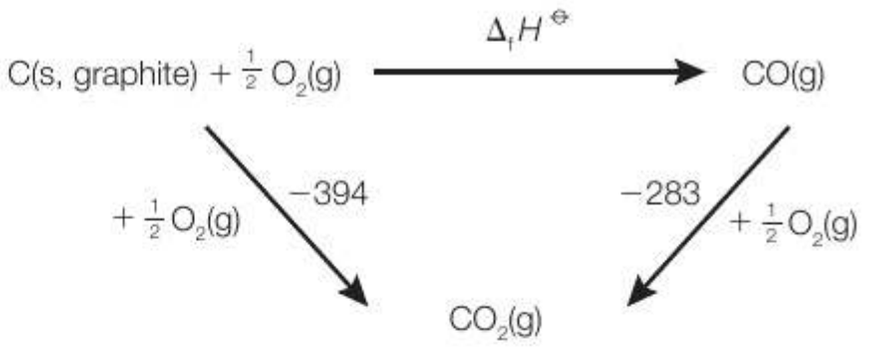
Adding the two equations gives:

$$C(s) + \tfrac{1}{2}O_2(g) \rightarrow CO(g) \qquad \Delta_c H^\ominus = -111\,kJ\,mol^{-1}$$

So, $\Delta_f H^\ominus[CO(g)] = -111\,kJ\,mol^{-1}$

METHOD 2: CONSTRUCT AN ENTHALPY CYCLE USING HESS'S LAW

An enthalpy cycle using Hess's Law is sometimes also called a Hess's Law cycle.

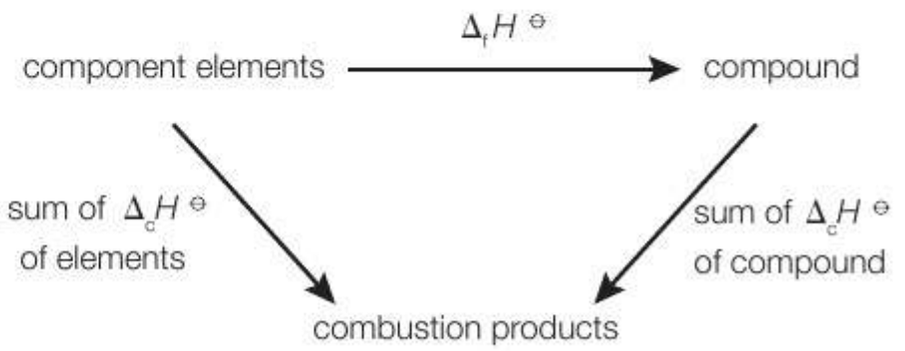


▲ **fig A** An enthalpy cycle using Hess's Law to calculate the standard enthalpy change of formation of carbon monoxide.

By Hess's law, $\Delta_f H^\ominus + (-283) = -394$

So, $\Delta_f H^\ominus = -394 - (-283) = -111\,kJ\,mol^{-1}$

In general terms, this is the enthalpy cycle used to calculate the enthalpy change of formation of a compound from the relevant enthalpy changes of combustion.



$\Delta_f H^\ominus$ = sum of $\Delta_c H^\ominus$ of elements − $\Delta_c H^\ominus$ of compound

▲ **fig B** An enthalpy cycle to calculate the enthalpy change of formation from enthalpy changes of combustion.

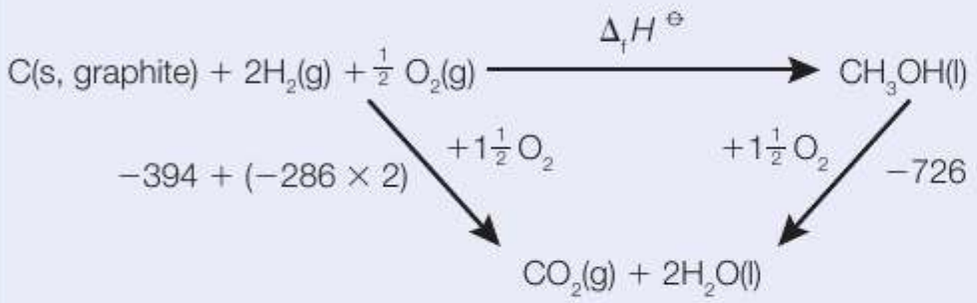
WORKED EXAMPLE 1

Calculate the enthalpy change of formation of methanol (CH_3OH), given the following enthalpy change of combustion data:

$\Delta_c H^\ominus[CH_3OH(l)] = -726\ kJ\ mol^{-1}$

$\Delta_c H^\ominus[C(s, graphite)] = -394\ kJ\ mol^{-1}$

$\Delta_c H^\ominus[H_2(g)] = -286\ kJ\ mol^{-1}$



▲ **fig C** An enthalpy cycle using Hess's Law to calculate the enthalpy change of formation of methanol.

$\Delta_f H^\ominus = -394 + (-286 \times 2) - (-726) = -240\ kJ\ mol^{-1}$

USING HESS'S LAW FOR OTHER REACTIONS

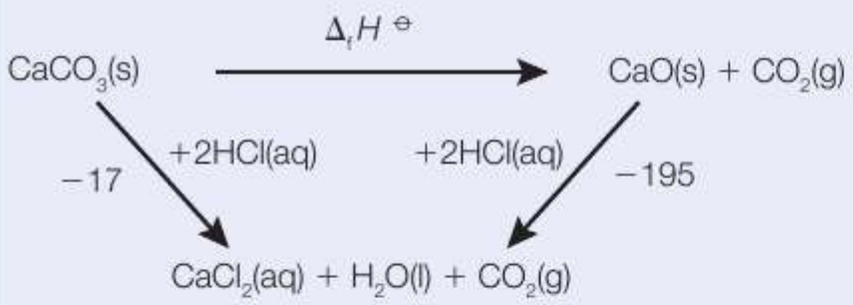
Hess's Law can be used to calculate the enthalpy change for many different types of reaction.

WORKED EXAMPLE 2

Calculate $\Delta_r H^\ominus$ for the thermal decomposition of calcium carbonate into calcium oxide and carbon dioxide, given the following data:

$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$
$\Delta_r H^\ominus = -17\ kJ\ mol^{-1}$

$CaO(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l)$ $\Delta_r H^\ominus = -195\ kJ\ mol^{-1}$



▲ **fig D** An enthalpy cycle using Hess's Law to calculate the enthalpy change of reaction for the thermal decomposition of calcium carbonate.

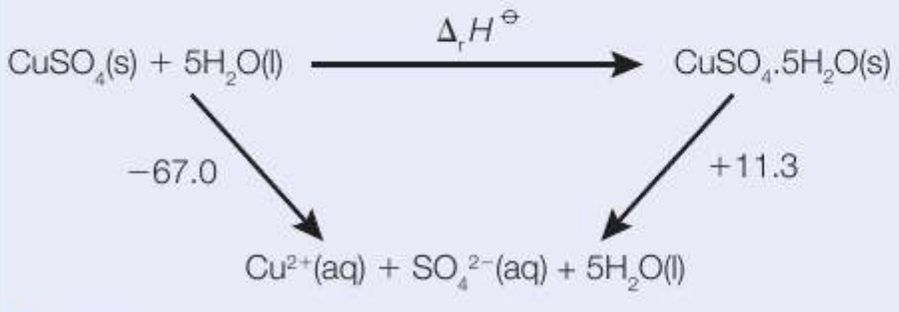
$\Delta_r H^\ominus = -17 - (-195) = +178\ kJ\ mol^{-1}$

WORKED EXAMPLE 3

Calculate $\Delta_r H^\ominus$ for the hydration of anhydrous copper(II) sulfate, given the following data:

$CuSO_4.5H_2O(s) + aq \rightarrow Cu^{2+}(aq) + SO_4^{2-}(aq) + 5H_2O(l)$
$\Delta_r H^\ominus = +11.3\ kJ\ mol^{-1}$

$CuSO_4(s) + aq \rightarrow Cu^{2+}(aq) + SO_4^{2-}(aq)$ $\Delta_r H^\ominus = -67.0\ kJ\ mol^{-1}$



▲ **fig E** An enthalpy cycle using Hess's Law to calculate the enthalpy change of reaction for the hydration of anhydrous copper(II) sulfate.

$\Delta_r H^\ominus = -67.0 - (+11.3) = -78.3\ kJ\ mol^{-1}$

CHECKPOINT

SKILLS PROBLEM SOLVING

1. (a) Write a chemical equation to represent the standard enthalpy change of formation of methane.

(b) Using the following data, calculate a value for the standard enthalpy change of formation of methane.

$\Delta_c H^\ominus[C(s, graphite)] = -394\ kJ\ mol^{-1}$

$\Delta_c H^\ominus[H_2(g)] = -286\ kJ\ mol^{-1}$

$\Delta_c H^\ominus[CH_4(g)] = -890\ kJ\ mol^{-1}$

2. Using the following data, calculate the standard enthalpy change for the reaction represented by the equation:

$AlCl_3(s) + 6H_2O(l) \rightarrow AlCl_3.6H_2O(s)$

$\Delta_f H^\ominus[AlCl_3(s)] = -704\ kJ\ mol^{-1}$

$\Delta_f H^\ominus[H_2O(l)] = -286\ kJ\ mol^{-1}$

$\Delta_f H^\ominus[AlCl_3.6H_2O(s)] = -2692\ kJ\ mol^{-1}$

3. The data shows some values of standard enthalpy changes of formation.

$\Delta_f H^\ominus[LiOH(s)] = -485\ kJ\ mol^{-1}$

$\Delta_f H^\ominus[H_2O(l)] = -286\ kJ\ mol^{-1}$

$\Delta_f H^\ominus[Li(s)] = 0.00\ kJ\ mol^{-1}$

(a) Why is the standard enthalpy change of formation of lithium quoted as zero?

(b) Write a chemical equation to represent the standard enthalpy change of formation of solid lithium hydroxide.

(c) Lithium reacts with water according to the following equation:

$2Li(s) + 2H_2O(l) \rightarrow 2Li^+(aq) + 2OH^-(aq) + H_2(g)$

Calculate the standard enthalpy change for this reaction. Use the data above and the following information:

$LiOH(s) + aq \rightarrow Li^+(aq) + OH^-(aq)$ $\Delta H = -21\ kJ\ mol^{-1}$

SUBJECT VOCABULARY

standard enthalpy change of formation the enthalpy change measured at 100 kPa and a specified temperature, usually 298 K, when one mole of a substance is formed from its elements in their standard states

Hess's Law the law states that the enthalpy change of a reaction is independent of the path taken in converting reactants into products, provided the initial and final conditions are the same in each case